

# STRAFE AIMING 101

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August 25, 2019

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**About me.** I'm a top FPS player<sup>1</sup> with more than 20 years of experience. I was recognized as the best HLDMM player for years, was consistently top 500 (4600 SR peak) in Overwatch from season 1 to season 7 on Tracer and Soldier, managed to get to 2650+ SR in duel in Quake Champion (top 10 online leaderboard) in a bit less than 4 months, and I'm obviously a KovaaK certified aim beast among many other useless stuff (like my relatively good movement and dodging skills). More importantly, I'm now a professional aim coach<sup>2</sup> for players of all skill level. In the past months, I've coached notable players who performed quite well (top 20 solo and/or duo) in the Fortnite World Cup, or in the more recent Solo and Trio Cash Cups. I've also coached a non neglectable amount of Apex pro or semi-pro players, and some Overwatch Contenders players.

**Why this guide?** Very simply, because almost nobody understands strafe aiming, plus the demand for such a guide is high. I actually teach this kind of stuff to people I coach privately, and I could perfectly sell this guide for \$10, but I believe that this knowledge is too important to be kept private. Any support would be greatly appreciated, as writing a guide with this level of depth is not easy and (very) time-intensive. Who knows, if there is enough support I might actually create new free content, but I'm not promising anything.

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<sup>1</sup>I should say ex, as I'm starting to be too old to compete with the new generation of extremely motivated and talented players

<sup>2</sup>this includes much more than just raw aim, as my previous guide on geometric positioning and the current one show. Aim is much more than "clicking heads"

## 1. INTRODUCTION

The goal of this guide is to succinctly explain the basics of strafe aiming. The latter is a generic term describing a collection of techniques involving both your mouse control and your movement skills. Without entering into too much details, strafe aiming is the “canonical” way to aim while moving with the purpose of maximizing your accuracy, that is, to deal the most damage regardless of being hit. Of course, because movement is involved, this guide will mostly interest players from fast movement speed (and fast acceleration) games, and relatively high time to kill games like Quake, Apex, and Overwatch. Before continuing, I would like to warn the reader that all of the definitions in this guide are mine. Some people might therefore disagree with the content of this guide. Now, to back up the content a bit, the point of view presented in this guide is supported by top lg duelers in quake (among many others), who are arguably the best players when it comes to movement skills and face to face aiming skills. More convincingly, I use these definitions on a daily basis with the people I coach, and the understanding of aiming skills they provide allows for a much deeper analysis of one’s strengths and weaknesses and hence for a better improvement rate. As for now, I don’t know any better framework to discuss this topic, or more generally, topics related to aiming skills.

The guide is organized into three parts. The first one introduces very general (yet important) concepts involved in aiming skills, before defining strafe aiming abstractly. The latter is then applied to define the so called “fundamental strafe aim forms”. Intermediate and advanced forms are then discussed in a relatively hard to get second part. The third and last section of this guide is very concrete and introduces different kind of routines, from beginner to advanced, to learn and to improve the different strafe aim forms defined previously. To finish this introduction, let me add that coupling aim with movement is a long and difficult process one shouldn’t expect to get good at quickly. In particular, I highly suggest people following my KovaaK guide to start training these strafe aim forms *seriously* once they are fully accomplished sub-intermediate.

## 2. WHAT IS STRAFE AIMING?

In everything that follows, suppose that you are playing a First Person Shooter with isotropic movements: the eight directions<sup>3</sup> are therefore equivalent (same speed, same acceleration, etc.). For simplicity, say that the max ground speed<sup>4</sup> is 10 units per second with respect to the (absolute) background map. Suppose that you’re also aiming with a tracking weapon with infinite many ammo and no spread nor recoil. Note that strafe aiming applies to click-timing weapons with fast enough fire rate as well (say, wingman or McCree), as tracking with such a weapon is a perfectly valid technique to click-time<sup>5</sup>.

**2.1. Some necessary terminology.** This part is just a dictionary of the skills involved when you aim. It isn’t made to be complete, nor subtle.

A very dumb definition of aim is *to put a crosshair at a place and at a time of your will*<sup>6</sup>. You obviously move your crosshair by moving your mouse on your mouse-pad in the physical space. Let’s name these skills the **mouse control** skills. They describe for example the smoothness of your mouse motion or your ability to make very clean straight lines or curves of different speed, or to make flicks of varied speed, length, and acceleration, along with your flick to track skills, precise micro-corrections, and so on. These skills are easily tested and built in aim trainers (as you stand still and can simulate any kind of movement), and if you have good mouse control in fps games you also have good mouse control in, say, paint or windows. The above definition of aim also contains an “on your will” part, which is clearly influenced by your awareness, and thus by the amount of information you’re able to process

<sup>3</sup>forward, backward, left, right, and the four diagonals

<sup>4</sup>you reach it once your character has accelerated for enough time, usually the acceleration is short and might even be almost instantaneous as in overwatch

<sup>5</sup>sometimes it even is the best way to maximize damage output

<sup>6</sup>this definition can be abused easily but it’s more than enough for the purpose of this guide

in real-time. Said differently, you first have to know where you want to aim at before deciding how to move your mouse to do so. These awareness skills are described by the **reading** and **reactivity** skills.

Quite generally, reading skills are describing your ability to be consciously aware of the exact position of your crosshair with respect to the target you want to aim at. They are closely related to your processing power, and this is why sleeping well is so important for aiming skills. Let's name a few of them:

- **precision reading**: the spatial resolution of your processing power, that is, how precise you are at discriminating the exact position of your crosshair with the position of the target you want to aim at.
- **fast-strafes reading**: the frequency of your processing power, that is, how fast you are at "refreshing" the crosshair position with respect to the enemy's position. This is a form of continuous reaction time. When your fast-strafes reading is bad, even though you might have a very good discrete visual reaction time (what you test on humanbenchmark), the scene appears blurry for much longer than your VRT. As the scene is blurry, you are lost and have no idea of the necessary corrections to apply, and either miss your target or hit it on luck, even with perfect mouse control skills.
- **movement reading**: this one is intuition/knowledge based, and is related to your ability to understand how movements at one time (your and the one of your target) affect the mouse motion you have to apply to hit your target at the next. Said differently, it describes how good you are at knowing how much your mouse should move at every apparent change of direction on your monitor. For example, flicking on a moving target involves movement reading skills, but so does tracking a target that changes direction while you don't.

Obviously, being target focused or crosshair focused, as explained in my KovaaK guide<sup>7</sup>, has a direct impact on these reading skills. Your crosshair type and its size too. When you play a scenario in an aim trainer and predict the movement of a bot for which the pattern is deterministic, you're only working on the mouse control skills and are neglecting the reading ones. When you aim at a target while standing still, you're trivializing movement reading too, as there is mostly two speed (10 ups<sup>8</sup> to the left and 10 ups to the right) to chose to move your mouse from. If you stop standing still, the combination of both your and the target movement adds a lot of complexity and variety in the apparent speed and acceleration, which makes movement reading hard. It is worth noting that these reading *skills* are trainable, and that there exists specific ways to focus on one or the other while you train.

Now, even if your mouse control skills were perfect and you managed to reach your genetic cap for the reading skills, you would still not have perfect aim. Suppose indeed, that you're aiming at a target that long strafes. Whenever the target changes direction, it takes a certain amount time for you to notice it because you can't have instantaneous reaction time. As soon as you notice the change of direction, the ideal way to correct your aim would be to perform an instant flick (infinite acceleration and infinite speed) followed by an aim-lock<sup>9</sup>. In layman's terms, your crosshair should ideally teleport instantly back on the target so as to not lose damage during the travel time of the correction. In practice, you don't have perfect mouse control and will often fail to hit a target that doesn't change direction because you don't have perfect smoothness, perfect precision, or aren't good at all speed. You can't make instant flicks either, and your post-flick tracking will necessarily be affected by the speed and acceleration of your flick. The time it takes for you to *react* as you miss, either by lack of mouse control, reading skills, or because your target just changed direction, is called the **reactivity**. Reactivity decomposes in two parts as explained above:

<sup>7</sup><https://www.dropbox.com/s/vaba3potfhf9jy1/>

<sup>8</sup>units per second

<sup>9</sup>you can't be smoother and more precise than that

- the **reactive part**: the time it takes for you to notice that you're missing, or that your target just changed direction. It is obviously deeply related to your reading skills, but this one as the merit of being quantifiable,
- the **correcting part**: the time it takes for you to flick back on target as soon as you realized that you are missing. This one is quantifiable too, as it is very easy to notice an acceleration (hence a correction) by looking at a video in slow-motion.

Let me add that it's hard to be smooth and reactive at once, as smoothness opposes itself to big accelerations. Depending on the game and on your play-style, being reactive might be better than being smooth. Quite generally, being smooth is better to build mouse-control skills, but slow people (say, VRT or reactivity test in footnote<sup>10</sup> above 200 ms) will certainly perform better with a reactive style rather than a smooth one.

**2.2. Definition and fundamental strafe aim forms.** This part defines strafe aiming before expliciting the most fundamental forms of strafe aim. To make the discussion simpler, suppose that the target you're aiming at is at medium range and is only allowed to use the left and right directions. A direction change for your enemy is therefore a left to right or a right to left direction change. By convention, the enemy is going to the left if he is pressing his left key, which means that he appears to move to the right on your monitor.

**Definition 2.1.** (Strafe Aiming) *To strafe aim is to change direction **whenever and only whenever** the target you're aiming at is changing direction.*

The first thing to remark is that strafe aiming does not constrain your mouse motion but your movement skills: you're **only allowed** and **forced** to change direction whenever the target you're aiming at does. Strafe aiming is as a result a reactive<sup>11</sup> aiming style, as you're not taking any opportunity to change direction on your own to make the enemy miss. That is, you're a priori (and a priori only) trying to maximize your damage output regardless of everything else. Now, as explained in the previous part, even with perfect reading skills and perfect mouse control, you cannot react instantly to a change of direction. This means that it is **impossible** to strafe aim a target that is changing direction too fast (what people often call short-dodging or ad-spamming). It is worth noting that with good reading skills, realizing that your enemy is short-dodging is relatively fast (of the order of 200 to 300ms), and that it is always possible to aim at the very middle of the direction changes to get a lot of free hits without having to lock the target at all. This also shows that short-dodging for too long is a very bad movement form. Now, because nobody has perfect reading skills, adding a short dodge to fake or to confuse your enemy is a very good strategy to make him miss (it isn't strafe aim then!).

Recall that there exists eight movement directions. At every change of direction of your enemy, by strafe aiming, you are forced to pick from 7 directions as you already move in one. As there exists 8 directions you can choose from for the former direction, there exists

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<sup>10</sup>For those who have KovaaK's FPS Aim Trainer and like to measure or compare their reaction time, here's EMAH7's reactivity test:

- (1) launch "tracking RT test" in KovaaK,
- (2) use OBS (studio, streamlab, ...) to record it in 120 FPS,
- (3) track the bot as smoothly as possible without predicting the (yet to come) direction change. When the bot changes direction, try to flick back on it as fast as possible while preserving a relatively smooth tracking then. Repeat this for 30 seconds to one minute,
- (4) stop recording and open the video using VLC. You're now going to count frames (this is done by pressing E successively, as E goes to the next frame) to measure your reactivity after every change of direction,
- (5) when counting, the first frame begins when your crosshair (ideally a small dot) is off the bot after a direction change. The last frame to be counted is at the middle of the correcting part, when your crosshair does not move with respect to the background (so before you accelerate it back on the target),
- (6) repeat this process for every direction change, and do the average,
- (7) your EMAH7 reactivity is your average number of frames  $\times$  8.33 ms.

<sup>11</sup>in the colloquial sense of the term, not related at all to the concept of reactivity above

$8 \times 7 = 56$  strafe aim forms for each change of direction of your enemy. That's quite a lot, and if we were rigorous, we should actually discriminate the strafe aim forms depending on the enemy's change of direction too<sup>12</sup>. Fortunately for us, assuming perfect reading skills (this is very strong), only the apparent motion of the enemy on your monitor matters. For example, an enemy changing direction from forward+left to forward+right with you reacting by a right to left change of direction is equivalent to an enemy changing direction from right to left and you reacting by forward+left to forward+right, at least for the mouse control skills.

Even with the equivalence above, there still exists 56 strafe aim forms to choose from. A natural way to classify them is to look at the **angle made by your change of direction**. The most fundamental strafe aim forms are the ones for which the angle is  $180^\circ$ , call them the *180-strafe aim forms*, that is:

- **mirroring**: you go right whenever (and only whenever obviously) your enemy goes left, and you go left whenever your enemy goes right, that is, you mirror the movement of your enemy. A lot of (bad) players confuse strafe aiming with mirroring. Do not listen to them, they don't understand shit about movement skills and aiming skills in general,
- **anti-mirroring**: you go left whenever your enemy goes left, and you go right whenever your enemy goes right, that is, you press the same movement keys as the target you aim at,
- **half-sideways anti-mirroring**<sup>13</sup>: there exists two of them, both involve diagonal directions. The first one is when you go forward+left whenever your enemy goes left, and you go backward+right whenever your enemy goes right. The second is the mirror image of this one, as in, you go backward+left whenever your enemy goes left, and you go forward+right.
- **half-sideways mirroring**<sup>14</sup>: there exists two of them, and they also involve diagonal directions. The first one is when you go forward+right whenever your enemy goes left, and you go backward+left whenever your enemy goes right. The second one is the mirror image of this one, as in, you go backward+right whenever your enemy goes left, and you go forward+left whenever your enemy goes right.
- **back and forth**: there exists two of them too, they are pretty much equivalent. The first one is when you go forward whenever the enemy goes left, and you go backward whenever the enemy goes right. The second one is obviously the opposite.

There exists therefore 8 180-strafe aim forms, that can be classified into 4 different types. They share different properties, pros, and cons, see next paragraph.

**2.3. Characteristics, pros, and cons of the 180-strafe aim forms.** First, let's recall that it is assumed that the max speed with respect to the background map is 10 ups. Of course, what matters for aim is the apparent speed of your target, that is, how you see it move on your monitor. This latter point of view is a bit complicated, as it depends of your distance from the target. An in between point of view is to speak in terms of relative speed and relative acceleration with respect to your reference frame. This is what I'm going to do in what follows. It is also assumed that the 8 directions are equivalent. Before continuing, it is very important to understand the *view-angle*, which is the direction at which your crosshair points. When you press forward and only forward, the view-angle is pointing in the same direction as where you move. When you move backward, it points at  $180^\circ$  from your movement direction, and when you use a diagonal, at either  $\pm 45^\circ$  or  $\pm 135^\circ$ .

Suppose that you strafe aim at an enemy that is going to the left (so on your right by convention). The latter now changes direction and is going to the right (so to your left). As you strafe aim, you're forced to change direction too. Remark something very interesting: if you stop moving the mouse as you change direction, your view-angle can either move toward the new direction of your target (say, if you now press left), or in the opposite one (say, if

<sup>12</sup>here we assumed that only a left to right change of direction happens

<sup>13</sup>abbreviation: hsw anti-mirroring

<sup>14</sup>abbreviation: hsw mirroring

you now press backward+right). While strafe aiming at a target, a change of direction is called *inward-directed* if the view-angle of your new direction is moving toward the new direction of your target. Otherwise, it is called *outward-directed*. An inward-directed change of direction is reducing the distance from your crosshair and the target during a change of direction. You therefore have to move the mouse less while correcting, hence reducing the correcting part of reactivity for free<sup>15</sup>. An outward-directed change of direction is increasing the distance from your crosshair and the target during a change of direction. You therefore have to move the mouse more while correcting, hence increasing the correcting part of reactivity.

A *strafe aim form* is called *inward-directed* if the two changes of direction are inward-directed, *outward-directed* when the two changes are outward-directed, and *mixed* otherwise. Then again, if we were rigorous, for a strafe aim form to be inward-directed, outward-directed, or mixed, explicitly depends of the nature of the change of direction of the target you're strafe aiming at. Now, assuming perfect reading skills, any change of direction of your target is equivalent (on your monitor) to a left-right change of direction. If you use this equivalence, be cautious that an inward-directed strafe aim form for a left-right moving enemy might become mixed or outward-directed.

- **mirroring:** inward-directed strafe aim form, makes movement reading easy as the relative speed is 0 ups. You don't have to move your mouse to aim as the relative speed is 0 ups outside of the changes of direction. During a change of direction the relative speed is at most 20 ups. Ideally, your correction is an instant flick with instant stop. This is hard to perform if the apparent size of the target is small. If you fail to correct, that's bad because you now have to beat static friction to micro-correct, which is annoying. If you don't move your mouse to aim, then you're not using your mouse control skills. The situation being symmetric, your opponent doesn't need to move his mouse either to hit you. If your mouse control is better than the one of your opponent, you should avoid mirroring as it ends up in traded damage<sup>16</sup>. Mirroring a target moving vertically is hard on reading, as you have to identify when to exactly stop the mouse horizontally and when to move it very gently in the good direction.
- **anti-mirroring:** outward-directed strafe aim form, makes movement reading relatively easy as the velocity is constantly 20 ups. Hard on mouse control as being smooth and precise at such a speed is difficult, especially if the target is close range. As you have to constantly move your mouse when you anti-mirror, you're using your mouse control skills a lot and are forcing the enemy to use them too, which leverages your edge in mouse control (if you have any). It beats static friction at all times, and thus make micro-correction easier. Interaction with vertical movement is easier than for mirroring. It's hard on reading to identify the difference of horizontal speed on a target moving vertically too. As the form is outward-directed and the velocity is 20 ups (the maximum), it encourages good reactivity.
- **hsw anti-mirroring:** outward-directed strafe aim form, makes movement reading relatively hard as the apparent velocity changes as you go further from your enemy or close the gap. This makes it harder than anti-mirroring on smoothness and mouse control (as you need to be quite precise with the horizontal acceleration of the mouse). The relative velocity is  $10 + 10 \cos(\pi/4) = 10 \times (1 + \sqrt{2}/2) = 17.07$  ups, which is a bit slower than anti-mirroring. It has the same advantage as anti-mirroring, and leverages your reading skills too. Using diagonals against someone who is moving left-right makes you easier to read for him than he is for you, as your absolute velocity on the background map is lower than his (reading skills are influenced by that). Most people are not smooth or precise enough to track diagonals properly though. Makes vertical angles easier, and makes you harder to hit from above as you move in a two dimensional plane thanks to the diagonal.

<sup>15</sup>at constant speed of the correcting flick

<sup>16</sup>which can be good if you have more hp than your opponent and can afford to get hit a lot

- **hsw mirroring:** inward-directed strafe aim form, the hardest on reading and mouse control as it requires very good precision, smoothness, and control of the accelerations. The relative speed is  $10(1 - \sqrt{2}/2) = 2.93$  ups, the target is almost motionless! This explains why this strafe aim form is so hard to read and to aim with<sup>17</sup>. The situation is almost symmetric, as it is very hard for your opponent to track you if he lacks precision and smoothness. Reading your movement is still slightly easier for him than it is for you to read him. It shares the same pros as hsw anti-mirroring with verticality, and is, by far, the best strafe-aim form to be precise at long range as it cancels most of the apparent motion of the target you aim at<sup>18</sup>. As this strafe aim form is inward directed, it maximizes the damage output. It does it better than any other form as it reduces the relative velocity of your opponent the most.
- **back and forth:** mixed strafe aim form, as changing direction from backward to forward is outward-directed while changing direction from forward to backward is inward-directed. Trivialize reading skills, as it is almost as standing still, makes you easy to hit, but allows for super precise micro-corrections. Is useful when you're safe, like in Overwatch when you are protected by a shield and aim at a distant widowmaker (say). It also makes you dodge people on your side for free. In a game with knockback on the tracking weapon like quake, pushing forward when someone falls and going backward as he is pushed up again due to the knockback (which is a change of direction) is the best way to maximize your damage output. Useless otherwise, and probably pointless to train for its own sake.

Remark that all of these 180-strafe aim forms share a major weakness: if the enemy is not moving globally because he goes left as much as he goes right<sup>19</sup>, neither are you. They are therefore globally static movement styles, and forbid you to reposition yourself on the map as you try to maximize your damage output. This is very bad, especially in a team game where you have to follow your leader, to secure a position, or to cover yourself from different enemies than the target you're aiming at. Moreover, because the diagonals are slower than left-right on the background map, you will eventually cross the crosshair of your opponent *for free*, giving him the opportunity to hit you without having to move his mouse. The same is obviously true for mirroring and anti-mirroring. Quite generally, you want to avoid giving free shots as much as possible, even when your goal is to maximize your damage output, it is very unlikely that you can afford to ignore "everything else".

### 3. INTERMEDIATE AND ADVANCED STRAFE AIM FORMS

This part is not going to be very exhaustive. Its purpose is to show what exists and how an a priori elementary aiming style such as strafe aim is not as trivial as it seems.

**3.1. Intermediate strafe aim forms.** The fundamental strafe-aim forms are all of the 180-strafe aim forms. Before defining the intermediate strafe aim forms, it is important to understand the concept of connection. Two strafe aim forms are *connected* when they share one prescription, that is, when they have a direction in common associated to the same enemy direction. Obviously, every fundamental strafe aim form is connected with itself, as you can always repeat them up to infinity. No two different 180-strafe aim forms are connected. When two strafe aim forms are connected, the change of direction they share is called a *connection*.

The building blocks of the intermediate strafe aim forms are (a subgroup of the) 90-strafe aim forms. More precisely, they are all of the the 90-hsw-forms connecting to the fundamental forms, that is:

- **forward triangle:** there exists two of them, one mirroring and one anti-mirroring version. For the mirroring version, you go forward+right whenever your enemy goes

<sup>17</sup>It is usual for (bad) players to believe that the mouse of a top strafe aimer using this form doesn't move. They don't appreciate how hard this form is because of that, sad how blind the average player is

<sup>18</sup>it beats static friction at all times as you're forced to continuously move the mouse

<sup>19</sup>symmetric movement style, which is awful, and is yet what most players do, even pros in the overwatch league

left and forward+left whenever your enemy goes right. This one is inward-directed. The anti-mirroring version, for which you go forward+left whenever your enemy goes left and forward+right whenever your enemy goes right, is outward-directed. Obviously in practice, you cannot use them up to infinity as you're pushing toward your enemy,

- **backward triangle:** there exists two of them too, one mirroring and one anti-mirroring version. For the mirror version, you go backward+right whenever your enemy goes left and backward+left whenever your enemy goes right. This one is inward-directed. The anti-mirroring version, for which you go backward+left whenever your enemy goes left and backward+right whenever your enemy goes right, is outward-directed. As for the forward triangle forms, you cannot use them up to infinity as you're disengaging from your enemy,
- **back and forth triangle:** There exists four of them, and all of them are mixed forms as you either only strafe to the right or only strafe to the left. When performed properly (this is hard), your global trajectory on the ground is a circle. You travel along this circle either in a clockwise or anti-clockwise direction, and always aim toward the center of it. Because strafing to the left is a mirror image of strafing to the right, I'm only defining the back and forth triangle forms where you strafe to the right for which you travel in a clockwise direction. For the first one, you go forward+right whenever your enemy goes to the left, and backward+right whenever the enemy goes to the right. For the second one, you go backward+right whenever your enemy goes to the left and forward+right whenever your enemy goes to the right. As mixed form, they are quite hard on precision, and because you move globally, you need to be good at large angles.

An *intermediate strafe aim form* is a connection, or a chain of connections of fundamental strafe aim forms along with the building blocks above. Let me give some examples:

- **back and forth left (or right) mirroring triangle:** there exists many different way to generate these strafe aim form by connections. A way to generate them is by connecting a backward mirroring triangle to a hsw-mirroring form, itself connected to a forward mirroring triangle. The result is a back and forth mirroring triangle. It is left directed or right directed depending on how you start it. They are obviously all inward-directed as they are mirroring strafe aim forms,
- **back and forth left (or right) anti-mirroring triangle:** same thing really, just use the anti-mirroring versions to generate them by connections. Obviously outward-directed,
- **back and forth top (or bottom) triangle:** they exist in many versions depending on which part of the triangle is inward-directed or outward-directed. Can you think of a way to generate them through connections?
- **rhombus**<sup>20</sup>: you have many of those too, depending on their orientation (either clockwise or anti-clockwise), and which part of the rhombus is inward-directed or outward-directed. Try to think of one example of each as an exercise, and how to generate them through connections.

Let me insist that listing all of these strafe aim forms is not intellectual masturbation at all: as movement reading skills are knowledge and intuition based, by understanding why seemingly equivalent shapes of your movement (say, different triangles or rhombi) require very different corrections, you're improving (slowly but surely) your strafe aim skills. This is quite obvious, as strafe aiming properly a left-right moving enemy depends heavily on the type of your strafe aim form (inward-directed vs outward-directed vs mixed, mirror vs anti-mirror versions, etc.). The more you think abstractly about how to move your mouse in one setting, the better you're going to perform in real situations.

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<sup>20</sup>in games with slightly slower side or diagonal speed, the shape is not a rotated square but a rhombus, hence the name



**3.2. Introduction to advanced strafe aim forms.** Before introducing some advanced strafe aim forms, let's summarize the previous parts. We first defined fundamental strafe aim forms as the collection of 180-strafe aim forms, that is, the forms for which the angle between the two changes of direction is  $180^\circ$ . The latter are completely static, which isn't great as they forbid repositioning. The concept of connection was then introduced. It is the key to generate the intermediate forms from a subgroup of the 90-strafe aim forms along with the fundamental ones. Notice that the intermediate strafe aim forms are mostly connections of hsw-forms (hsw anti-mirror, backward triangle, rhombus, ...). Those give the first examples of forms that are non-static, as some of those globally push forward or backward, to the right or to the left, or even circle around your enemy. The latter are yet directed<sup>21</sup> hsw-forms. Being hsw is also not optimal, as it never maximizes the relative speed seen from the perspective of your enemy nor the absolute speed on the background map, which makes you slightly easier to hit than your enemy is. The only two non-hsw forms that are considered intermediate are the fundamental mirroring and anti-mirroring ones.

Now, quite generally, a good movement has a **purpose**: it either maximizes damage output (say, a strafe aim form), it either minimizes damage taken (say, a dodging pattern), it either repositions yourself on the map to gain an advantage, for example, by abusing *geometric positioning* (see my extensive guide<sup>22</sup> on the subject). Obviously, no movement is doing everything at once. Some are stronger at one of these points and weaker at others. The same is true for the advanced strafe aim forms: some of them are actually quite good at minimizing damage taken while still dealing a considerable amount of damage. An example of one such is the following:

- **back to forward-hsw**: there exists right and left versions. They are all 45-strafe aim forms (hence triangles). A right version of the back to forward-hsw is when you go backward whenever your enemy goes right and you go forward+right whenever your enemy goes left. This form has to be used at relatively close range when your enemy is swapping direction relatively quickly<sup>23</sup>, and when he is on the right of your monitor. Here's why: when you're close range and you go backward when your enemy goes right, it isn't easy for him to aim at you as you significantly increase the angle to the left from his perspective. Your enemy's crosshair is likely to be late on you, which gives him an incentive to aim with movement by swapping direction to the left. As your enemy goes left, you push forward+right (to his left then). As the range is close, and because you push, you're actually turning his change of direction in an outward-directed one! Moreover, your enemy has to turn the mouse much more than you do (as this form is inward-directed), which is hard for him and easy for you.

If you understood the previous example properly, notice that the change of direction of your enemy is *predictable*, yet you still strafe-aim. This is typical of advanced strafe aim forms: you need a very good timing to execute them properly, and a good intuition of the typical tendencies of most players to react to their lack of mouse of control with their movement. Of course, good dodgers are not easily predictable as they react to your reactions, and so on. Still, here's another example:

- **forward-hsw to mirroring**: there exists right and left versions of those too, and all are 45-strafe aim forms. The right forward-hsw to left mirroring form is when you go forward+right whenever your enemy goes left, and left whenever your enemy goes right. This one is called a right version because you push to the right, and mirror to the left. It is one of the best form to push someone, as aiming at you hsw-mirroring is heavy on precision for your enemy. As a result, the latter will have the tendency to change direction (and to go to his right) to increase the relative speed between you and him for which his mouse control is likely better. As he does so, you react by going left and mirror him. Then again, if you predict the change of direction properly (or even

<sup>21</sup>in the sense that they only move on one side

<sup>22</sup><https://www.dropbox.com/s/aif0jy1prxe0rjm/>

<sup>23</sup>you can actually predict the change of direction of your enemy, as this movement form will increase the tendency of your enemy to mirror you

initiate it) your enemy is likely to be late on you as you go left, and he likely cannot afford to go left either because it is too hard for his mouse control. In particular, if executed properly, this form is at all times inward-directed. Of course, the good reaction for your enemy is to break the left-right pattern, and to use diagonals or backward only here, but that would lead us to the dodging skills realm, way outside of the scope of this guide (and way harder to explain abstractly).

There's obviously much more 45° strafe aim forms. All of them are fundamental blocks of advanced forms. All the "rectangular"<sup>24</sup> 90°-strafe aim forms are fundamental blocks too, here's one example:

- **back to right mirroring rectangle:** inward-directed strafe aim form, you go backward whenever the enemy goes right, and right whenever he goes left. These forms are very useful to reposition yourself or to mess with the aim of your opponent if he is changing direction relatively quickly, and here's why: as you go backward while the enemy goes right, he is moving his mouse to the left to track a 10 ups (relative) motion. During the change of direction, he starts moving his mouse to the right. You now react to his change of direction and move to the right, that is, to his left: he has to flick to the left and to stop his flick exactly on your body because you're now mirroring. If he fails the flick, he is likely to change direction (because it's easier to correct by beating static friction, that is, by anti-mirroring in this case), and so should you. For this exact same remark, the most useful rectangular forms are the mirroring ones. Even then, you should use those with a purpose in mind (repositioning yourself, getting safe, trying to fuck the aim of your opponent to then engage into another strafe aim form or to dodge, ...).

Let me insist once again that the remarks above are actually quite general and are not proper to strafe aim forms: **your movement has to have a purpose**, so you can couple it with the game-sense positioning and mechanical skills. Think of geometric positioning: if you know exactly what position allows you hit for free but are relatively far from it, you can always connect different strafe aim forms so as to move globally toward it. The same is true for minimizing damage, and this is regardless of dodging skills.

To complete the fundamental blocks of the advanced forms, we would have to add all of the 135° forms too. What are we omitting? Actually nothing: the building blocks are exactly all of the 56 forms we counted in the beginning of the guide. *An advanced strafe aim form is a connection, or a chain of connections, of these 56 forms.* That's quite a lot, isn't it?

#### 4. ROUTINES TO IMPROVE YOUR STRAFE AIM

The goal of this part is to present some routines, from beginner to advanced skill level, to improve your strafe aim and more generally your movement aim. They require KovaaK's FPS Aim Trainer, which is still the best aim trainer on the market. The conventions used are the same as in my KovaaK guide. That said, let me add that the next routines are mostly experimental, as most of the scenarios were created for the purpose of this guide (based on my intuition of what would constitute a good workout) and have not been tested by anyone yet. I would also like to thank patys, one of the most prolific scenario creator in KovaaK, who helped me to create these scenarios from scratch. Do not hesitate to play his scenarios<sup>25</sup>, as most of them are very good (the standing-still ones too).

Before presenting the routines, let me explain the different movement scoring system in KovaaK, and why they are not great:

- **Distance Traveled Scoring:** you earn points by the distance traveled, regardless of the movement directions. This gives a strong incentive to "long dodge"<sup>26</sup> rather than

<sup>24</sup>then again, if the side speed is slower than the backward speed or the forward speed, the shape is a rectangle and not a square

<sup>25</sup>quite some are based on my ideas, but don't remind it to him or he is going to rage. Still, he implemented them nicely

<sup>26</sup>to keep moving in the same direction after a change of direction of your target

to strafe aim. While long dodging is a perfectly legitimate way to move, it is advanced, situational, and not strafe aim. Moreover, because points are awarded regardless of the direction you move toward, you can cheese scoring by only using back and forth, which is obviously a really bad way to train your movement aim,

- **Movement-Based Scoring**<sup>27</sup>: a bar representing movement points fills as you press the left or the right direction. It fills up according to some rules defined by the scenario. You “bank” points by changing direction (say, from right to left), but if you wait for too long, the points cap as the bar can be full. If you change direction too fast (when the color of what fills the bar is not green) you get less points too. This movement system is good to learn how to aim while short-dodging, but it’s bad to learn advanced strafe aim forms. It also gives an incentive to change direction too early, and it makes you more static. You can also cheese by strafing toward a wall, as the system still counts points even though you’re not moving.

The best system to train strafe aim would be a conditional movement scoring system. Here’s how it would look like:

- **Conditional Movement Scoring**: you only earn points by following a strafe aim prescription, say, by strafing left whenever and only whenever the bot goes left, and forward+right whenever and only whenever the bot goes right. With such a system you would actually have the perfect scoring system<sup>28</sup> to train strafe aim forms.

If any aim trainer developer reads this guide, you have some good idea for free here (hi kovaak, it’s been 1 year since I gave you this idea).

The conclusion is that movement scoring systems in KovaakK (and everywhere else really) are bad right now. This means that score is not *that* relevant in most of the training maps below. For the routines to work efficiently, it is assumed that you are of good faith and are not trying to cheese (as always, but here even more). In any case, this is only the very beginning of training movement and aim together in aim trainers. I’m very optimistic that in the near future, there will be much better and diverse scenarios to train strafe aiming and long dodging. I’m also very optimistic that aim trainers are going to be improved drastically and will be unavoidable. Who knows, the dream of training against a machine learning bot being your dodge-nemesis might not be that far.

**4.1. Beginner routine.** Before starting your beginner routine, go in the options, bind the aimbot to your fire key, and launch Horizontal Bounce Dodge in free play. Try to change direction in synch with the bot. Feel free to experiment different strafe aim forms (all the fundamental ones, and some intermediate too). This helps your movement reading, as the aimbot has perfect mouse control, it shows you exactly how to move the mouse for a given strafe aim form. The beginner routine is the following:

*Beginner Routine:*

- **Pole Long Dodge 5 min.:** learn how to correct during a change of direction and to deal with the acceleration. Train left to right and right to left change of direction, or the diagonal ones. You can play with the range too, circle the bot or whatever you can think of,
- **Horizontal Bounce Dodge** (Vertical Bounce Dodge) **5 min.:** train anti-mirroring, hsw-mirroring, and hsw-antimirroring forms on HBD. On VBD, try to change direction at every bounce (this is a change of direction even if it’s vertical!) and at least once in between,
- **Rhombus Dodge 10 min.:** use the fundamental strafe aim forms and try to maximize score, feel free to experiment what happens at different angles of the rhombus,
- **Rhombus Vertical Dodge** (XY Tracking Dodge) **10 min.:** use the fundamental strafe aim forms and try to maximize score, feel free to experiment what happens at

<sup>27</sup>that’s kovaak name, and it’s awful

<sup>28</sup>you would have to tweak it a bit, like, awarding some points up to 200ms after a direction change of the bot to forbid prediction, etc.

different ranges. You'll naturally press forward as the bot goes up, and backward as it goes down, try to invert that too. On XY Tracking dodge, just do as best as you can, gl,

- **Close LS Easy Dodge** (Close FS Easy Dodge) **10 min.:** try to maximize score with everything you have learned so far.

Once you feel good enough on this routine, you can try the intermediate one.

**4.2. Intermediate routine.** As before, try to use the intermediate strafe aim forms in free play while aimbotting. If you want to experiment with some advanced forms, I suggest you play Close LS Easy Dodge, as you can circle around the bot and push it. Quite generally, do not hesitate to aimbot in freeplay to gain some intuition about a strafe aim form.

*Intermediate Routine:*

- **Pole Med Dodge** **5 min.:** train the fundamental strafe aim forms (even if the bot doesn't move) and change direction before the bar fills completely,
- **Pole Long Dodge** **5 min.:** train the intermediate strafe aim forms (forward and backward triangles, back and forth triangles, rhombus, ...) and change direction before the bar fills completely,
- **Horizontal Bounce Dodge** (Vertical Bounce Dodge) **10 min.:** train all of the intermediate forms you know (the fundamental forms are part of the intermediate ones!)
- **Rhombus Dodge** (Rhombus Vertical Dodge) **10 min.:** use the intermediate strafe aim forms and try to maximize score, feel free to experiment what happens at different angles
- **Close LS Easy Dodge** (Close LS Easy Dodge High Ground, Close LS Easy Dodge Low Ground, Close LS Dodge) **10 min.:** try to maximize score with everything you learned,
- **LGC3 Reborn Easy** (Close FS Easy Dodge, XY Tracking Dodge) **10 min.:** try to maximize score with everything you have learned so far.

**4.3. Advanced routine.** Same as before, you can always aimbot and test some stuff. As for scoring, I have a very good heuristic here for you to know if you're a fully advanced strafe aimer or not: try to get 95% of nick's score, the god of strafe aim (and I'm not even joking).

*Advanced Routine:*

- **Pole Short Dodge** **5 min.:** train the fundamental strafe aim forms (even if the bot doesn't move) and change direction before the bar fills completely,
- **Pole Med Dodge** **5 min.:** train the intermediate strafe aim forms (forward and backward triangles, back and forth triangles, rhombus, ...) and change direction before the bar fills completely,
- **Pole Long dodge** **5 min.:** train all of the forms you know and try to connect them to make an advanced strafe aim form you could use in real game,
- **Rhombus Dodge** (Rhombus Vertical Dodge, Close LS Dodge) **10 min.:** train the intermediate and advanced strafe aim forms, feel free to experiment what happens at different angles of the rhombus,
- **Close FS Easy Dodge** (Close FS Easy Dodge High Ground, Close FS Easy Dodge Low Ground) **10 min.:** try to maximum score
- **LGC3 Reborn** (LGC3 Reborn Varied, Air Dodge, XY Tracking Dodge Hard, Close FS Dodge) **10 min.:** try to maximum score

To conclude, let me just say that muscle memory is a meme, but that's for another guide, maybe another day.